

Tailoring Machine Learning Weather Predictions for Local Impacts

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Research Goal and Overview

- Show the feasibility of finetuning weather forecasts for specific outcomes using minimal compute power
- Use a simple MLP model to correct both a data driven ML forecast (Pangu) and a traditional NWP forecast (IFS HRES)
- Demonstrate the method using 2 meter temperature and 10 meter wind speed across a variety of regions and lead times
- 2m Temp Improvements range from XX-XX
- 10m Wind Speed Improvements range from XX-XX

Key Contribution: Demonstrating a simple method to significantly improve forecast accuracy using limited computational power.

Previous that Post-Processing NWP Forecasts with Neural Networks is Effective

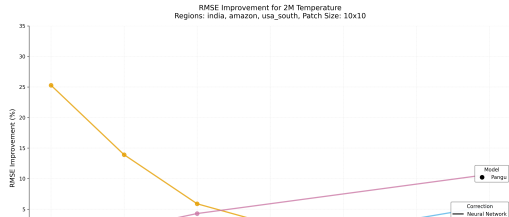
- citehanDeepLearningMethod2021 trained a U-Net to correct IFS Forecasts in Northern China
 - Improvements in 2m Temp RMSE go range from 30% to 12% for lead times 1 to 7 days
 - Improvements in 10m Wind Speed RMSE Range from 25% to 20% for lead times 1 to 7 days
- citearaujoPostProcessingAirTemperature2022 test using both a CNN and MLP to further improve post processing model GRIDPP over Norway
- Uses RMSE error instead of MSE but I find a larger percent improvement for both 2m temperature and 10m wind speed

Method

- Uses Pangu and IFS HRES forecasts with lead times from 1 to 7 days
 - Target Pangu predictions using ERA5 and IFS HRES predictions using its $t=0$ initialization
- Filter to a single variable and lead time, in a particular region
- Train 128x5 MLP network that takes in Pangu/IFS forecast outputs and targets MSE compared to target ERA5/IFS HRES data
 - Train using a single forecast per day for years 2018-2021
 - Test using daily forecasts in 2022

Headline Result: Simple MLP Seems Effective at Correcting Forecasts

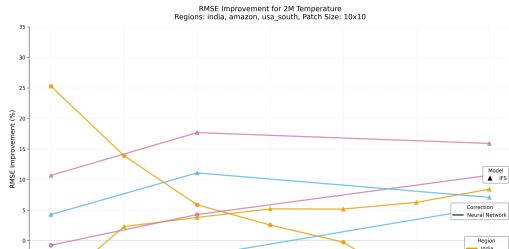
- Simple MLP method works across variety of regions and climates
- Training a correction model for a 10x10 degree region takes about 2.5 minutes on an M3 max MacBook Pro



`finetuning/pangu/lead_time/multi_region`

Headline Result: Simple MLP Seems Effective at Correcting Forecasts

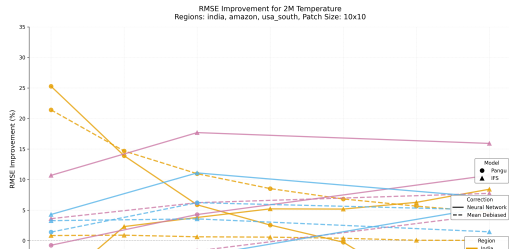
- Simple MLP method works across variety of regions and climates
- IFS HRES forecasts can be improved a similar amount using the same method



`finetuning/pangu/lead_time/multi_region`

Headline Result: Simple MLP Seems Effective at Correcting Forecasts

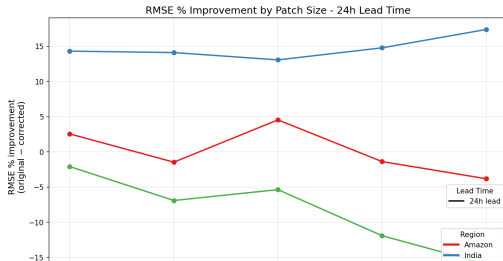
- Simple MLP method works across variety of regions and climates
- MLP greatly outperforms just subtracting mean forecast error for each pixel



`finetuning/pangu/lead_time/multi_region`

Results are Inflexible to Training Region Size

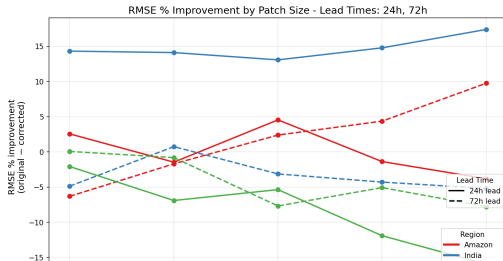
- RMSE of central 2x2 region depending on how much of the surrounding area is included in training



finetuning/pangu/subregion/subregion_r

Results are Inflexible to Training Region Size

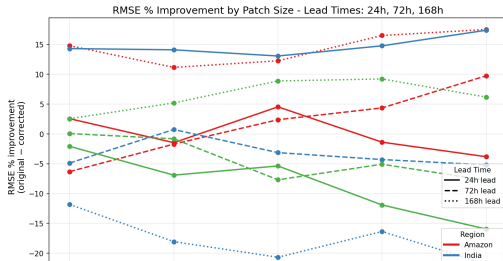
- RMSE of central 2x2 region depending on how much of the surrounding area is included in training



finetuning/pangu/subregion/subregion_r

Results are Inflexible to Training Region Size

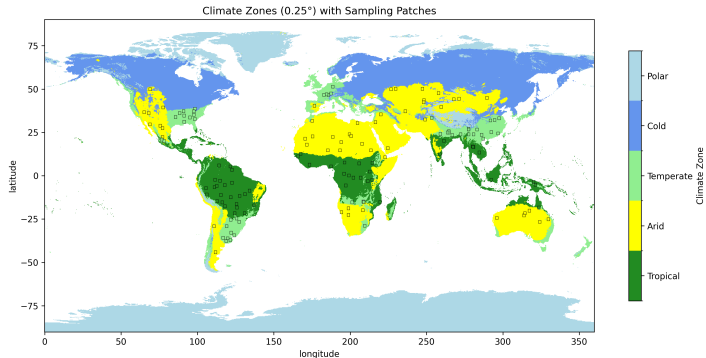
- RMSE of central 2x2 region depending on how much of the surrounding area is included in training



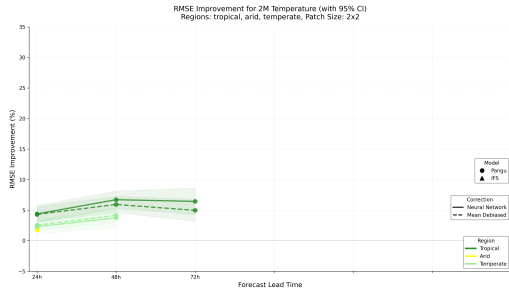
`finetuning/pangu/subregion/subregion_r`

How to Explore Heterogeneity by Climate Region

- Define tropical, temperate, and arid pixels on an 0.25 degree grid using broad Koppen-Geiger classes
- Train fine-tuning model on randomly sampled 2x2 degree patches



Heterogeneity by Climate Region: 2m Temperature

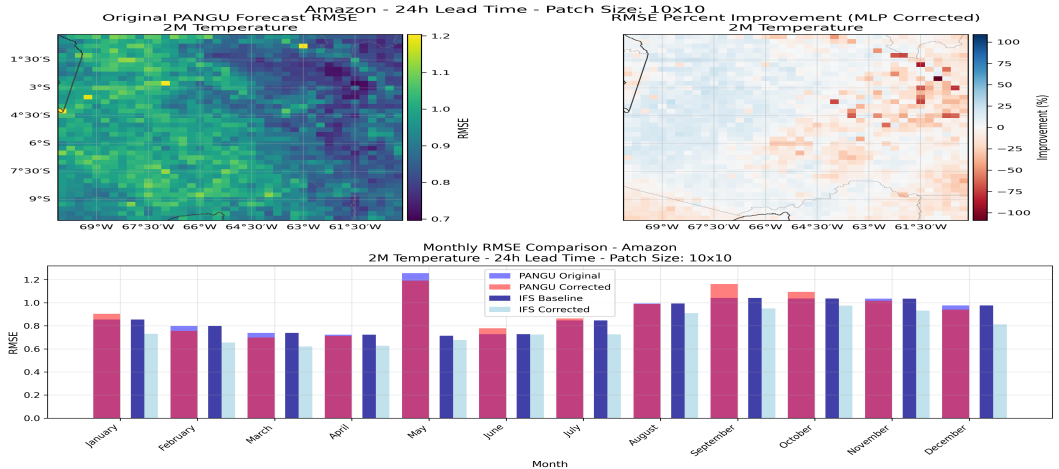


(a) 2m Temperature

finetuning/pangu/lead_time/multi_region

(b) Wind Speed

Amazon Temperature



Amazon Wind Speed

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British Columbia Temperature

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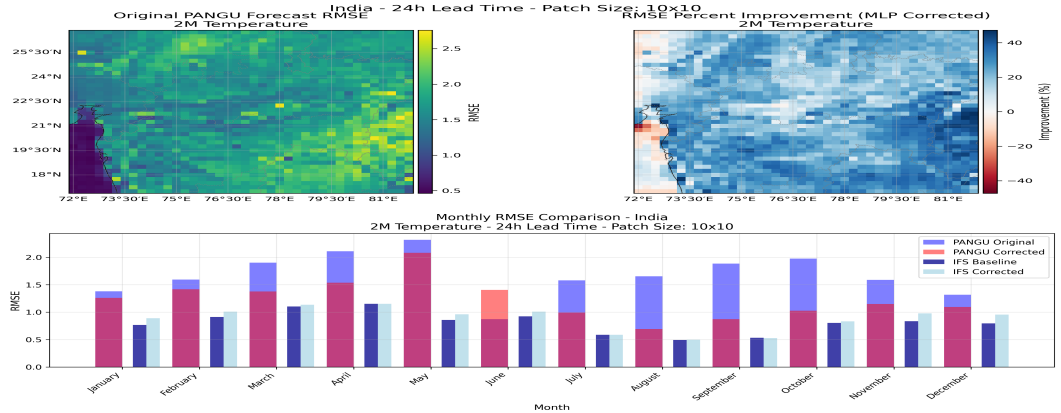
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British Columbia Wind Speed

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India Temperature

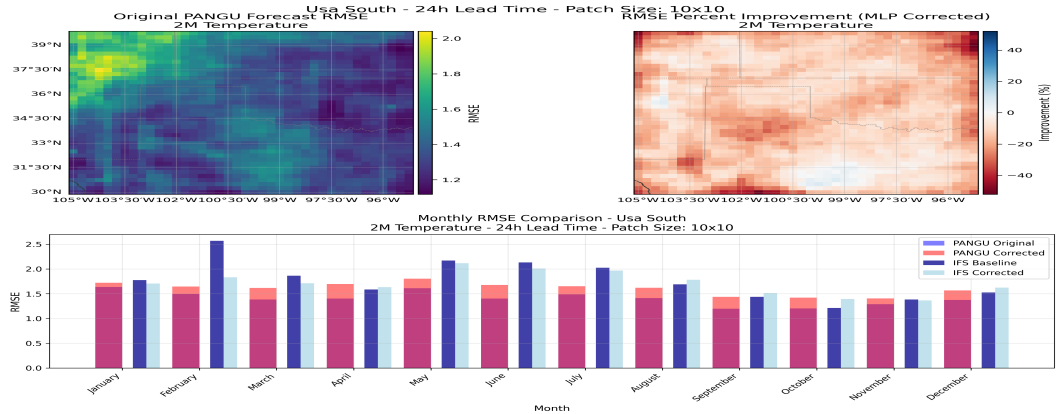


India Wind Speed

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USA South West Temperature



USA South West Wind Speed

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finetuning/pangu/time_series/usa_south/10x10/rmse_monthly_10m_wind_speed_t
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What I'm Actively Working on

- Rewriting the databuild to be able to take advantage of the new WeatherbenchX package and also be able to run on a server and be more flexible
- Implementing a U-Net model for fine-tuning
 - Early results suggest it significantly ups compute time without much improved performance
- Implement ANO Correction (subtracting mean error) method to compare to MLP correction